Multiclass classification - Softmax regression

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1 K-class classification with softmax

Given a dataset consisting of (N) observations for a multi-class classification task:

- Let $x_i \in \mathbb{R}^P, i = 1, \ldots, N$ denote the input vector for observation (i). (If the first element of (x_i) is the intercept term, then $x_i = \left[1, x_{i1}, \ldots, x_{i,P-1}\right]^T$.)
- Suppose there are K classes,

$$\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_K.$$

- The class label of observation (i) is represented using a **one-hot encoded** vector y_i. If observation i belongs to class $\mathcal{C}_k$, then:

$$y_i = \begin{bmatrix} y_{i1} \\ ... \\ y_{ik} \\ ... \\ y_{iK} \end{bmatrix} = \begin{bmatrix} 0 \\ ... \\ 1 \\ ... \\ 0 \end{bmatrix},$$

  $y_{ik} = 1$ and $y_{ij} = 0$ for all $j \neq k$. $y_{i1} + y_{i2} + y_{i3} + ... + y_{iK} = 1$.

We model the probability that a sample i with feature vector x_i belongs to class k using softmax function as follows:

$$p(y_i \in \mathcal{C}_k | x_i; W) = \pi_{ik} = \frac{e^{w_k^T x_i}}{\sum_{k'=1}^{K} e^{w_{k'}^T x_i}}.$$

Softmax function is a general form of logistic regression.

Define the weight matrix W:

$$W = \begin{bmatrix} w_1^T \\ ... \\ w_K^T \end{bmatrix}, \tag{1}$$

in which $w_k \in \mathbb{R}^P$ is the weight vector associated with class k.

Likelihood of observation x_i, y_i is given by:

$$p(y_i \mid x_i; W) = \prod_{k=1}^{K} (\pi_{ik})^{y_{ik}}. \tag{2}$$

Hence, the likelihood for all observations is given by:

$$p(y|X; W) = \prod_{i=1}^{N} p(y_i|x_i) \tag{3}$$

$$= \prod_{i=1}^{N} \prod_{k=1}^{K} (\pi_{ik})^{y_{ik}}. \tag{4}$$

Maximum likelihood problem: Finding optimal W such that maximizing $p(y|X; W)$.

Logarithm function transforms a multiplication to an addition, which is easier to optimize. **Maximum log-likelihood problem** is given by:

$$\text{max.} \sum_{i=1}^{N} \sum_{k=1}^{K} \log \left((\pi_{ik})^{y_{ik}} \right) \tag{5}$$

$$\sum_{i=1}^{N} \sum_{k=1}^{K} y_{ik} \log \pi_{ik} \tag{6}$$

$$= -\sum_{i=1}^{N} CE(y_i, \pi_i) \tag{7}$$

where $CE(.,.)$ is the cross-entropy between y_i and $\pi_i = [\pi_{i1}, \ldots, \pi_{iK}]^T$.

The above problem is a convex.

2 Example

Students may generate an artificial dataset and use the CVXPY library to obtain the optimal solution to the softmax problem (see this reference `https://machinelearningcoban.com/2017/02/17/softmax/` for example on artificial dataset generation).